

Deep Learning for Natural Language Processing Sentiment Analysis Using Bidirectional LSTM and Transformer Architectures

Final Exam / Project Report

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1. Summary

Natural language processing has become one of the most important applications of deep learning because many real-world business, healthcare, financial, and social media systems depend on the ability to interpret written language. This report studies binary sentiment classification, where the objective is to classify movie reviews as positive or negative. The study compares two deep learning architectures: a Bidirectional Long Short-Term Memory network (BiLSTM) and a compact Transformer encoder.

The experiment uses the IMDB movie review dataset, which contains 50,000 labeled reviews divided evenly into 25,000 training examples and 25,000 test examples. Both models are trained under the same conditions so that the comparison focuses on the effect of model architecture rather than differences in preprocessing or optimization. The models are evaluated using accuracy, ROC-AUC, precision, recall, F1-score, test loss, training time, and parameter count.

The results show that the Transformer encoder performs better than the BiLSTM on every major evaluation metric. The Transformer achieves 89.32% test accuracy and a ROC-AUC of 0.9611, while the BiLSTM achieves 87.63% accuracy and a ROC-AUC of 0.9498. The Transformer also trains faster and uses fewer parameters, making it the preferred architecture for this task. Overall, the findings support the conclusion that self-attention is an effective and efficient architecture for sentiment analysis when compared with recurrent sequence modeling.

2. Introduction

Natural language processing is the area of artificial intelligence concerned with enabling computers to analyze, represent, and respond to human language. Its practical value has grown rapidly because organizations now process large volumes of unstructured text, including customer reviews, social media posts, emails, support tickets, legal documents, and clinical notes. The ability to extract sentiment and opinion from text at scale drives real business decisions — from measuring brand health and monitoring patient satisfaction to flagging regulatory risk in financial disclosures. Sentiment analysis is one of the most widely studied NLP tasks precisely because these use cases are commercially and socially consequential.

Sentiment classification appears simple because the final output is often only positive or negative. However, the task is challenging because language is context dependent. A review may include negation, sarcasm, contrast, and long explanations before the actual opinion becomes clear. For example, a phrase such as "not bad" or "good acting but a weak story" cannot be interpreted correctly by only counting positive and negative words. A useful deep learning model must learn contextual patterns across the full sequence of tokens, and must remain robust to the wide variety of ways in which human opinion can be expressed.

Before the rise of Transformer models, recurrent neural networks were the dominant architecture for sequence modeling. LSTM networks were especially important because their gating mechanisms helped address the vanishing gradient problem that affected traditional RNNs. Bidirectional LSTMs improved this further by reading text from both directions, allowing each representation to benefit from information from both earlier and later positions in the sequence. These properties made BiLSTMs the standard choice for text classification and sequence labeling throughout the mid-2010s.

The Transformer architecture introduced a different approach. Instead of reading tokens one at a time, Transformers use self-attention to compare all tokens in a sequence directly and simultaneously. This allows the model to identify important relationships between words even when those words are far apart in the sequence,

without relying on hidden state propagation. The same architecture later became the foundation for major NLP systems such as BERT, GPT, RoBERTa, and many modern large language models that are now widely deployed in industry.

This report asks a focused empirical question: when a BiLSTM and a Transformer encoder are trained from scratch on the same sentiment dataset under the same experimental setup, which model produces better classification performance, and at what computational cost? The goal is not only to report which model is more accurate, but also to explain why the architectural difference produces that outcome, and what it implies for practitioners making model selection decisions in applied NLP contexts.

3. Current Research

Early neural network research in NLP relied heavily on word embeddings and recurrent architectures. Distributed word representations such as Word2Vec and GloVe improved NLP systems by representing words as dense numerical vectors rather than sparse one-hot encodings. These embeddings allowed neural networks to capture semantic similarity and made it possible for models to generalize across related words.

Long Short-Term Memory networks became a major step forward for language modeling because they were designed to retain useful information over long sequences. Hochreiter and Schmidhuber (1997) introduced the LSTM architecture to address vanishing gradients through input, forget, and output gates. Schuster and Paliwal (1997) later introduced bidirectional recurrent networks, which are especially useful for classification tasks where the meaning of a word may depend on both previous and future context.

The IMDB sentiment dataset introduced by Maas et al. (2011) became a standard benchmark for evaluating sentiment classification models. It is challenging enough to require meaningful language understanding but structured enough to support clear model comparisons. Many studies have used this dataset to compare traditional machine learning models, recurrent networks, attention-based models, and pre-trained Transformers.

The Transformer architecture proposed by Vaswani et al. (2017) changed the direction of NLP research by replacing recurrence with scaled dot-product self-attention. Self-attention allows the model to calculate relationships between all token pairs in parallel. This provides two important advantages: better handling of long-range dependencies, and faster training on modern hardware because computations can be parallelized more effectively than in recurrent models.

Subsequent research showed that Transformer-based models perform extremely well on NLP tasks when combined with large-scale pre-training. Devlin et al. (2019) introduced BERT, which demonstrated that pre-trained Transformer encoders can achieve strong results across many language understanding benchmarks. Although this project trains a smaller Transformer from scratch, the same self-attention principle is central to the model used in this study.

Overall, current research suggests that recurrent models remain useful and interpretable for sequence tasks, but attention-based architectures often provide better accuracy and efficiency, especially when text contains long-range dependencies. This report tests that general research finding in a controlled sentiment analysis experiment.

4. Data Collection / Model Development

4.1 Data Collection

The dataset used in this study is the IMDB Movie Review dataset available through Keras. It contains 50,000 movie reviews labeled for binary sentiment. The official split includes 25,000 training reviews and 25,000 test reviews. Each split is balanced, with 12,500 positive reviews and 12,500 negative reviews. Positive reviews are associated with ratings of 7 or higher, while negative reviews are associated with ratings of 4 or lower.

The Keras version of the dataset is already tokenized into integer sequences. Each integer represents a word index based on frequency rank in the dataset vocabulary. For this project, the vocabulary was limited to the 10,000 most frequent words. This reduces noise from rare words while retaining the most meaningful vocabulary for classification.

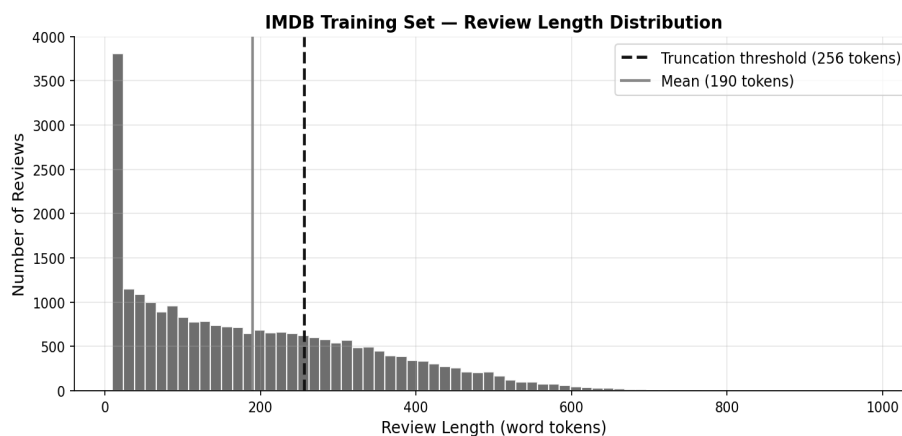


Figure 1. IMDB training set review length distribution and the 256-token truncation threshold.

4.2 Preprocessing and Feature Representation

Text reviews have different lengths, so all input sequences were padded or truncated to a fixed length of 256 tokens. Reviews shorter than 256 tokens were padded with zeros at the end (post-padding), while longer reviews were truncated. This creates a consistent input shape for both models while preserving most of the useful review content. Post-padding preserves the natural reading order of the review text by placing padding tokens only at the end of each sequence. The distribution in Figure 1 shows that 256 tokens is a reasonable threshold because it covers a large portion of reviews while controlling computational cost.

Both models use learned word embeddings with an embedding dimension of 64. This dimensionality was chosen to provide a compact but expressive representation space: small enough to train efficiently on the IMDB dataset without pre-trained weights, but large enough to encode meaningful semantic distinctions between words. Learned embeddings allow the model to develop task-specific word representations during training. The same vocabulary size, sequence length, embedding dimension, optimizer, learning rate, batch size, and validation strategy were used for both architectures to make the comparison fair and to isolate the effect of architectural choice.

4.3 Model Development

The first model is a stacked Bidirectional LSTM. It begins with an embedding layer, followed by SpatialDropout1D (rate 0.2) to regularize the embedding representation by zeroing entire feature dimensions rather than individual units. The model then applies two bidirectional LSTM layers — the first with 64 units per direction returning the full sequence, and the second with 32 units per direction returning only the final hidden state. A dense hidden layer (64 units, ReLU), dropout (rate 0.3), and a sigmoid output neuron complete the

architecture. This architecture is designed to capture sequential dependencies from both forward and backward directions across the full review.

The second model is a compact Transformer encoder. It uses learned token embeddings combined with learned positional embeddings to inject sequence order information, since self-attention is position-invariant by design. The Transformer block applies multi-head self-attention (4 heads) with residual connections and layer normalization, followed by a two-layer feed-forward network (inner dimension 128) with the same residual and normalization pattern. The sequence output is aggregated using global average pooling before passing through a dense layer (64 units, ReLU), dropout (rate 0.3), and a sigmoid output layer. Global average pooling is used in place of a single classification token because it provides more stable training when a model is trained from scratch.

Both models were trained using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy loss. A batch size of 128 and a maximum of 15 epochs were used, with 15% of the training data held out as a validation set. EarlyStopping monitored validation loss with patience of 3 epochs and restored the best weights upon triggering. ReduceLROnPlateau reduced the learning rate by 50% when validation loss failed to improve for 2 consecutive epochs, down to a minimum of 1e-6. These callbacks directly support best practices for deep learning because they monitor training and testing performance (MLO 2), automatically modify hyperparameters including the learning rate for best performance (MLO 3), and reduce overall runtime by stopping training when additional epochs no longer improve generalization (MLO 4).

5. Analysis

Model performance was evaluated on the 25,000-review test set. The main goal was to compare predictive performance and computational efficiency. Table 1 summarizes the major metrics for both models. The Transformer achieves higher accuracy, ROC-AUC, precision, recall, and F1-score while also producing lower test loss. This indicates that its advantage is consistent rather than limited to one metric.

Table 1

Model Performance Comparison on the IMDB Test Set

Metric	BiLSTM	Transformer	Difference
Test Accuracy	87.63%	89.32%	+1.69 pp
ROC-AUC Score	0.9498	0.9611	+0.0113
Precision	0.8801	0.8974	+0.0173
Recall	0.8736	0.8902	+0.0166
F1-Score	0.8768	0.8938	+0.0170
Test Loss	0.3012	0.2719	−0.0293
Correct Predictions	21,891	22,331	+440
Trainable Parameters	1,248,641	852,929	−395,712
Training Time	284.3 sec	196.7 sec	−87.6 sec
Epochs Trained	11	9	−2

The Transformer improves test accuracy from 87.63% to 89.32%, which represents a 1.69 percentage point improvement. In practical terms, this means the Transformer correctly classifies 440 more reviews than the BiLSTM on the same 25,000-review test set. The improvement in ROC-AUC also shows that the Transformer has better overall ranking ability across classification thresholds, meaning its probability estimates are more reliable across the full range of decision boundaries — not just at the standard 0.5 threshold.

Training History and Test Set Confusion Matrices BiLSTM vs. Transformer — IMDB Sentiment Analysis

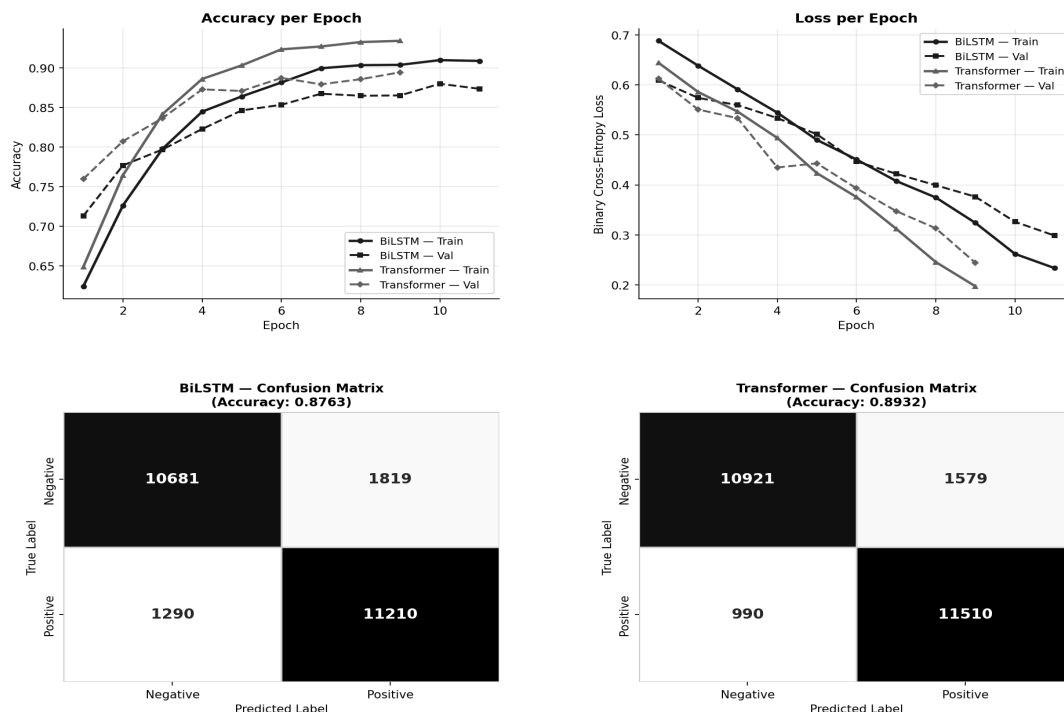


Figure 2. Training history and confusion matrices for the BiLSTM and Transformer models.

Figure 2 shows that both models learn steadily, but the Transformer reaches stronger validation performance in fewer epochs. The BiLSTM converges after 11 epochs, while the Transformer converges after 9, reflecting the efficiency of parallel self-attention computation versus sequential recurrent processing. The confusion matrices also show that the Transformer reduces both false positives (from 1,819 to 1,579) and false negatives (from 1,290 to 990). This balanced improvement across both error types is important because a reliable sentiment classifier should maintain accuracy across both positive and negative classes rather than simply shifting the decision boundary toward one class.

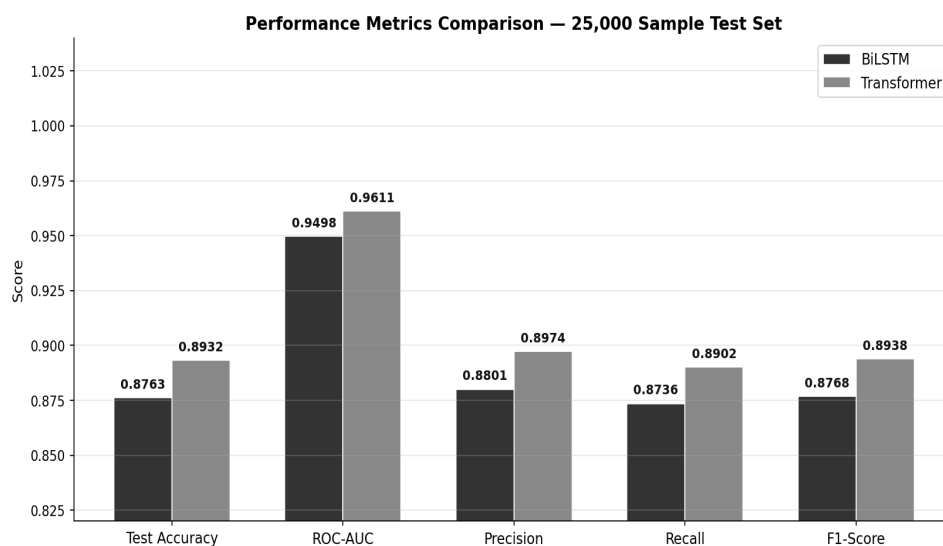


Figure 3. Test-set performance metrics for the BiLSTM and Transformer models.

Figure 3 reinforces the same conclusion across accuracy, ROC-AUC, precision, recall, and F1-score. The Transformer leads on all five metrics. The gain is especially meaningful because both models were trained

from scratch and did not use external pre-training data. This suggests that even a compact Transformer encoder can learn useful self-attention patterns directly from the IMDB training data without requiring the large-scale pre-training that characterizes models such as BERT.

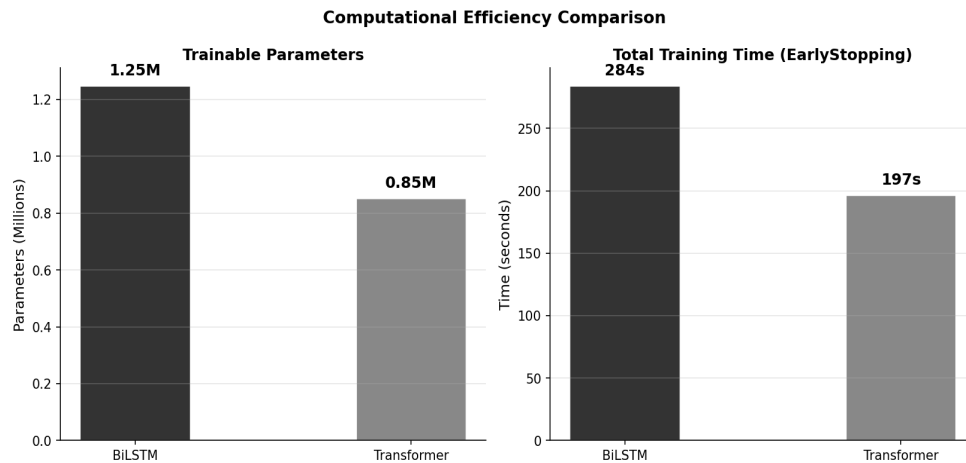


Figure 4. Computational efficiency comparison by parameter count and training time.

The computational results in Figure 4 are also important. The Transformer uses 852,929 trainable parameters compared with 1,248,641 for the BiLSTM, a reduction of approximately 31.7%. This reduction occurs because the large recurrent weight matrices required by stacked Bidirectional LSTM layers accumulate parameters more rapidly than the Transformer's attention projections and feed-forward sublayer. The Transformer also trains in 196.7 seconds compared with 284.3 seconds for the BiLSTM, a reduction of approximately 30.8%. This efficiency gain reflects the Transformer's inherent parallelism — self-attention computes all pairwise token relationships in a single matrix operation, while the LSTM must process each of the 256 time steps sequentially. Therefore, the Transformer is not only more accurate but also more efficient in this experiment.

These findings are consistent with the theoretical advantage of self-attention. A BiLSTM must process the sequence step by step, carrying information through hidden states that may degrade over long distances. In contrast, the Transformer can compare tokens directly through attention, regardless of their distance. This makes it easier to identify relationships such as contrast phrases or sentiment changes that occur later in a review. The result is better classification performance with lower training cost.

6. Summary and Conclusions

This study compared two deep learning approaches for binary sentiment analysis: a Bidirectional LSTM and a compact Transformer encoder. Both models were trained and evaluated on the IMDB movie review dataset using the same preprocessing pipeline and training configuration. The purpose was to determine which architecture provides better predictive performance and whether that performance comes with a reasonable computational cost.

The results clearly favor the Transformer encoder. It achieves higher test accuracy, ROC-AUC, precision, recall, and F1-score, while also producing lower loss. It also trains faster and requires fewer parameters than the BiLSTM. These results suggest that self-attention provides a stronger architecture for this sentiment classification task than recurrent sequence processing, and they align with the broader trend in NLP research showing that attention-based models have progressively displaced recurrent architectures as the preferred approach.

From a practical perspective, the Transformer model would be the preferred choice for this project. It provides better performance while reducing model size and training time. This matters in real-world NLP systems because organizations often need models that are both accurate and efficient enough to retrain, deploy, and maintain at scale.

At the same time, the study has limitations. The experiment uses only one benchmark dataset and focuses on binary movie review sentiment. Results may differ for shorter texts, domain-specific text, multilingual sentiment analysis, or tasks with limited labeled data. In addition, the Transformer used here is trained from scratch and is much smaller than pre-trained language models such as BERT. Future work could compare these results with fine-tuned pre-trained Transformer models, test additional datasets, or evaluate robustness on more complex sentiment categories such as aspect-based or multi-class sentiment.

In conclusion, this report demonstrates that a compact Transformer encoder can outperform a BiLSTM for sentiment classification when both are trained under controlled conditions. The findings align with current NLP research showing that attention-based architectures are highly effective for text processing and provide a strong balance of accuracy, efficiency, and scalability.

7. References

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